

Mine Closure Planning

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- ▶ Initial, Progressive and Final Mine closure Planning and its components
- ▶ Auditing; Legal and Financial Aspects.



Introduction

- ▶ Mine closure means steps taken for reclamation, restoration measures taken in respect of a mine or part thereof commencing from close of mining or processing operation in a mine or part thereof
- ▶ Every mine shall have a Mine Closure Plan which shall be of two types:
 - Progressive Mine Closure Plan
 - Final Mine Closure Plan
- ▶ The Plans have to be approved by IBM or State government. The plans shall cover protective, reclamation and rehabilitation activities
- ▶ The lease holder shall furnish financial assurance for implementation of plan, which could be in the form of letter of credit from a Scheduled Bank, Security Bond etc.

Issues Related to the Mine Closure

- ▶ Environmental Protection and Reclamation
- ▶ Disposal of Asset
- ▶ Legal Framework

1. Mine closure should be **integral** to the whole of mine life plan.
2. A **risk-based approach** to planning should reduce both cost and uncertainty.
3. **Closure plans** should be developed to reflect the status of the project or operation.
4. **Closure planning** is required to ensure that closure is technically, economically and socially feasible.

The dynamic nature of closure planning requires **regular and critical review** to reflect changing circumstances

- ▶ Activities related to closure planning often include:
 - ▶ preparation of detailed drawings of disturbed landscape,
 - ▶ compilation of baseline information,
 - ▶ discussions with regulators and stakeholders on end land use considerations,
 - ▶ crafting of supporting research programs, and preparation of budgets and schedules.
- ▶ For a mine nearing closure, the closure plan takes the form of a decommissioning plan, and includes details and selection of mitigative technologies (especially for acid rock drainage) and other specific reclamation and closure activities.
- ▶ Progressive Mine Closure Plan would include various land use activities to be done continuously and sequentially during the entire period of the mining operations, whereas the Final Mine Closure activities would start towards the end of mine life, and may continue even after the reserves are exhausted and/or mining is restored to acceptable level.

OBJECTIVES OF MINE CLOSURE PLANNING

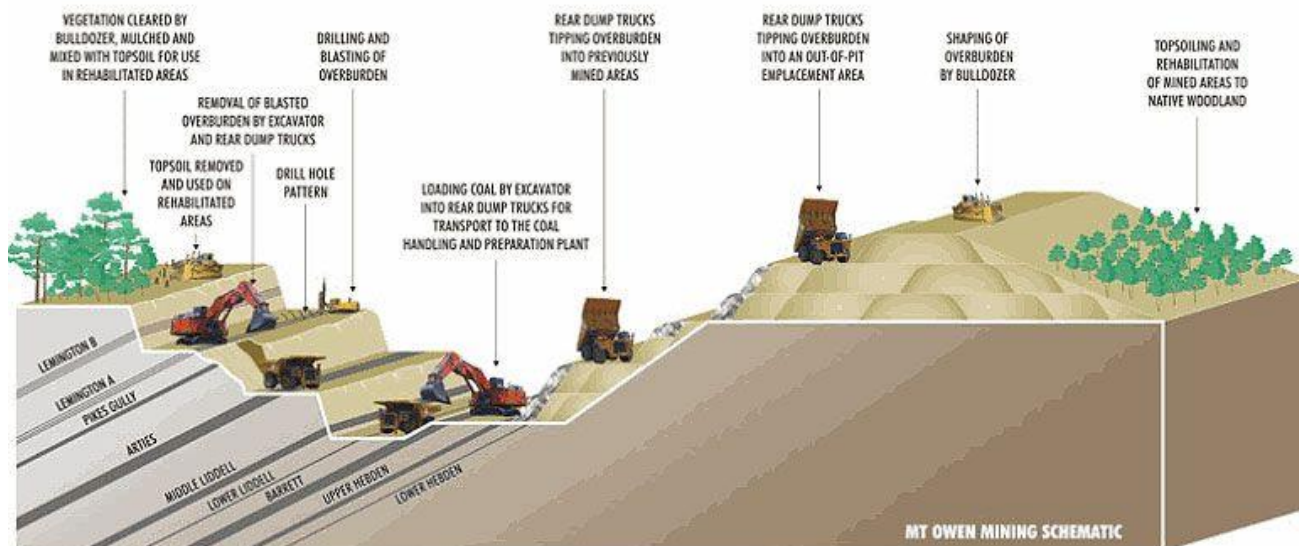
The development of a Closure Plan needs to take into account both the legal requirements and the unique environmental, economic and social aspects of the operation. In developing the Closure Plan, the following four key objectives should be kept in mind:

- ▶ To protect the environment and public health and safety by using safe and responsible closure practices;
- ▶ To reduce or eliminate environmental effects once the mine operations are ceased;
- ▶ To establish conditions which are consistent with the pre-determined land-use objectives at the end of mining operations;
- ▶ To reduce the need for long-term monitoring and maintenance by establishing effective physical and chemical stability of disturbed areas.

So, in essence, the objective of mine closure planning is “To ensure that the process of closure occurs in an orderly, cost-effective and timely manner”

Guidelines for preparation of Mine Closure Plan, IBM

- 1. The Central Government vide Notification No. GSR 329 (E) dated 10.04.2003 and No. GSR 330 (E) dated 10.04.2003 amended the Mineral Concession Rules, 1960 and Mineral Conservation and Development Rules, 1988 respectively. As per these amendments all the existing mining lessees are required to submit the "Progressive Mine Closure Plan" along with prescribed financial sureties within 180 days from date of notification. Further, the mining lessee is required to submit "Final Mines Closure Plan" one year prior to the proposed closure of the mine. In the notification it has been enumerated that the "Progressive Closure Plan" and "Final Closure Plan" should be in the format and as per the guidelines issued by the Indian Bureau of Mines.



- ▶ 2. Mine closure encompasses rehabilitation process as an ongoing programme designed to **restore physical, chemical and biological quality disturbed by the mining to a level acceptable to all concerned**. It must aim at leaving the area in such a way that rehabilitation **does not become a burden to the society after mining operation is over**. It must also aim to create as self-sustained ecosystem.
- ▶ 3. Mine closure operation is a continuous series of activities starting from day one of the initiation of mining project. Therefore, **progressive mine closure** plan will be an additional chapter in the present mining plan and will be **reviewed every five years in the Scheme of Mining**. **As progressive mine closure is a continuous series of activities, it is obvious that the proposals of scientific mining have had included most of the activities to be included in the progressive mine closure plan.**
- ▶ 4. **Final mine closure plan as per statute, shall be considered to have its approval at least nine months before the date of proposed closure of mine.** This period of nine months is reckoned as preparatory period for final mine closure operations. Therefore, all proposals for activities which have to be carried out after production of mineral from the mine or mining is ceased, shall be included in the final mine closure plan.

CLOSURE PLANS

- ▶ Closure plans should be developed to reflect the status of the project or operation. At least two types of closure plan will be required through the life of a mine; a Conceptual Closure Plan (project phase) and the Actual Closure Plan (operations phase):
 - A Conceptual Closure Plan for use during feasibility, development and detailed design; and
 - An Actual Closure Plan for use during construction, operation and post-operation
- ▶ **Conceptual Closure Plan:** A Conceptual Closure Plan identifies the key objectives for mine closure to guide project development and design. It should include broad land use objectives and indicative closure costs. (This does not preclude land use objectives being varied during the mine life to reflect changes in both knowledge and technology).
- ▶ **Actual Closure Plan:** Closure planning includes a commitment to progressive restoration and detailed plan development and implementation. A number of subsidiary plans need to be developed as the Closure Plan evolves. These typically include: a rehabilitation plan, a decommissioning plan and a maintenance and monitoring plan.

- ▶ **Rehabilitation plan:** A key component of the Closure Plan is a commitment to progressive rehabilitation. In conjunction with an active research and trial programmes, this may assist in minimizing ongoing contamination and reduce final costs by confirming or modifying completion criteria and demonstrating that they can be met. **Progressive rehabilitation allows best use of available personnel and equipment and should assist in minimizing required security deposits.**
- ▶ **Decommissioning plan:** As a detailed component of the Closure Plan, a decommissioning plan should be developed towards the final stages of an operation. **(As the exact date for ceasing production is rarely known, it is suggested that the decommissioning plan be developed 2 to 4 years prior to estimated cessation).** Once established it should be updated annually. The decommissioning plan includes such things as: details of the demolition and removal or burial of all structures not required for other uses; removal, remediation or encapsulation of contaminated materials; and the procedures for making safe and sealing, openings to underground workings.
- ▶ **Maintenance and monitoring plan:** The last aspect of the Closure Plan is performance monitoring, which should be designed to demonstrate that the completion criteria have been met. This period should also plan for remedial action where monitoring demonstrates completion criteria are unlikely to be met. If progressive rehabilitation has been successful, with stabilisation and revegetation meeting completion criteria this last phase of closure may be shortened. It is, however, unlikely to be less than 5 years in duration.

Content of Closure Plans (Indian Views)

1. Introduction:

- ▶ The name of the lessee, the location and extent of lease area, the type of lease area (forest, non-forest etc), the present land use pattern, the method of mining and mineral processing operations, should be given.
- ▶ 1.1 *Reasons for closure* : The reasons for closure of mining operations in relation to exhaustion of mineral, lack of demand, uneconomic operations, natural calamity, directives from statutory organisation or court etc. should be specified.
- ▶ 1.2 *Statutory obligations* : The legal obligations, if any which the lessee is bound to implement like special conditions imposed while execution of lease deed, approval of mining plan, directives issued by the Indian Bureau of Mines, conditions imposed by the Ministry of Environment and Forests, State of Central Pollution Control Board or by any other organisation describing the nature of conditions and compliance position thereof should be indicated here.
- ▶ 1.3 *Closure plan preparation* : The names and addresses of the applicant and recognised qualified person who prepared the Mine Closure Plan and the name of the existing agency should be furnished. A copy of the resolution of the Board of Directors or any other appropriate administrative authority as the case may be on the decision of closure of mine should be submitted.

2. Mine Description:

- ▶ 2.1 **Geology**: Briefly describe the topography and general geology indicating rock types available, the chemical constituents of the rocks / minerals including toxic elements if any, at the mine site.
- ▶ 2.2 **Reserves** : Indicate the mineral reserves available category wise in the lease area estimated in the last mining plan / mining scheme approved along with the balance mineral reserves at the proposed mine closure including its quality available (for final mine closure plan only).
- ▶ 2.3 **Mining Method** : Describe in brief the mining method followed to win the mineral, extent of mechanisation , mining machinery deployed, production level etc.
- ▶ 2.4 **Mineral Beneficiation** : Describe in brief the mineral beneficiation practice if any indicating the process description in short. Indicate discharge details of any tailings / middlings and their disposal / utilisation practice followed.

3. Review of Implementation of Mining Plan / Scheme of Mining including five years Progressive Closure Plan upto final closure of mine:

- ▶ Indicate in detail the various proposals committed with special emphasis on the proposals for protection of environment in the approved Mining Plan / Scheme of Mining including five years Progressive Closure Plan upto the closure of mine vis-a-vis their status of implementation. Highlight the areas, which might have been contaminated by mining activities and type of contaminants that might be found there. The reasons for deviation from the proposals if any with corrective measures taken should also be given.

4.Closure Plan :

- ▶ **4.1 Mined-Out Land** : Describe the proposals to be implemented for reclamation and rehabilitation of mined-out land including the manner in which the actual site of the pit will be restored for future use. The proposals should be supported with relevant plans and sections depicting the method of land restoration / reclamation / rehabilitation.
- ▶ **4.2 Water Quality Management** : Describe in detail the existing surface and ground water bodies available in the lease areas and the measures to be taken for protection of the same including control of erosion, sedimentation, siltation, water treatment, diversion of water courses , if any, measures for protection of contamination of ground water from leaching etc. Quantity and quality of surface water bodies should also be indicated and corrective measures proposed to meet the water quality conforming the permissible limits should also be described. Report of hydrological study carried out in the area may also be submitted. The water balance chart should be given. If there is potential of Acid Mine Drainage the treatment method should be given.
- ▶ **4.3 Air Quality Management** : Describe the existing air quality status. The corrective measures to be taken for prevention of pollution of air should be described.
- ▶ **4.4. Waste Management** : Describe the type, quality and quantity of overburden, mineral reject etc. available and their disposal practice. If no utilisation of waste material is proposed, the manner in which the waste material will be stabilised should be described. The protective measures to be taken for prevention of siltation, erosion and dust generation from these waste material should also be described. If toxic and hazardous elements are present in the waste material the protective measures to be taken for prevention of their dispersal in the air environment, leaching in the surface and ground water etc, should be described.
- ▶ **4.5 Top Soil Management** : The top soil available at the site and its utilisation should be described.

- ▶ **4.6 Tailing Dam Management :** The steps to be taken for protection and stability of tailing dam, stabilisation of tailing material and its utilisation, periodic desilting, measures to prevent water pollution from tailings etc., arrangement for surplus water overflow alongwith detail design, structural stability studies, the embankment seepage loss into the receiving environment and ground water contaminant if any should be given.
- ▶ **4.7 Infrastructure :** The existing infrastructural facilities available such as roads, aerial ropeways, conveyer belts, railways, power lines, buildings & structures, water treatment plant, transport, water supply sources in the area etc. and their future utilisation should be evaluated on case to case basis. If retained, the measures to be taken for their physical stability and maintenance should be described. If decommissioning proposed, dismantling and disposal of building structures, support facilities and other infrastructure like electric transmission line, water line, gas pipeline, water works, sewer line, telephone cables, underground tanks, transportation infrastructure like roads, rails, bridges, culverts etc., electrical equipments and infrastructures like electric cables, transformers to be described in connection with restoring land for further use.
- ▶ **4.8 Disposal of Mining Machinery :** The decommissioning of mining machineries and their possible post mining utilisation, if any, to be described.
- ▶ **4.9 Safety and Security :** Explain the safety measures implemented to prevent access to surface openings, excavations etc., and arrangements proposed during the mine abandonment plan and upto the site being opened for general public should be described.
- ▶ **4.10 Disaster Management and Risk Assessment :** This should deal with action plan for high risk accidents like landslides, subsidence flood, inundation in underground mines, fire, seismic activities, tailing dam failure etc. and emergency plan proposed for quick evacuation, ameliorative measures to be taken etc. The capability of lessee to meet such eventualities and the assistance to be required from the local authority should also be described.
- ▶ **4.11 Care and maintenance during temporary discontinuance :** For every five yearly review (as given in the mining scheme), an emergency plan for the situation of temporary discontinuance or incomplete programme due to court order or due to statutory requirements or any other unforeseen circumstances, should include a plan indicating measures of care, maintenance and monitoring of status of unplanned discontinued mining operations expected to re-open in near future. This should detail item wise status monitoring and maintenance with periodicity and objective.

5. Economic Repercussions of closure of mine and manpower retrenchments :

- ▶ Manpower retrenchment, compensation to be given, socio-economic repercussions and remedial measures consequent to the closure of mines should be described, specifically stating the following.
- ▶ 5.1 Number of local residents employed in the mine, status of the continuation of family occupation and scope of joining the occupation back.
- ▶ 5.2 Compensation given or to be given to the employees connecting with sustenance of himself and their family members.
- ▶ 5.3 Satellite occupations connected to the mining industry - number of persons engaged therein - continuance of such business after mine closes.
- ▶ 5.4 Continued engagement of employees in the rehabilitated status of mining lease area and any other remnant activities.
- ▶ 5.5 Envisaged repercussions on the expectation of the society around due to closure of mine.

6. Time Scheduling for abandonment : The details of time schedule of all abandonment operations as proposed in para 4 should be described here. The manpower and other resources required for completion of proposed job should be described. The schedule of such operations should also be supplemented by PERT (Programme Evaluation & Review Technique), Bar chart etc.

- ▶ **7. Abandonment Cost** : Cost to be estimated based on the activities required for implementing the protective and rehabilitation measures including their maintenance and monitoring programme.
- ▶ **8. Financial Assurance** : The financial assurance can be submitted in different forms as stated in Rule 23(F)(2) of Mineral Conservation and Development (amendment) Rules, 2003. In the mine closure plan, the manner in which financial assurance has been submitted and its particulars have to be indicated. For Model Bank Guarantee Form (**for surface mine-Rs 9 lakhs per hectare and for underground mine Rs 1.5 lakhs per hectare**).
- ▶ **9. Certificate** : The above mentioned actions have been taken to be stated clearly in the mine closure plan. A certificate duly signed by the lessee to the effect that said closure plan complies all statutory rules, regulations, orders made by the Central or State Government, statutory organisations, court etc. have been taken into consideration and wherever any specific permission is required the lessee will approach the concerned authorities. The lessee should also give an undertaking to the effect that all the measures proposed in this closure plan will be implemented in a time bound manner as proposed.
- ▶ **10. Plans, Sections etc.** : The chapter 1,2,3 and 4 should be supported with Plans and Sections. The Closure Plan may also be submitted depicting photographs, satellite images on compact disc etc. wherever possible.

Assessing Closure Risk

- ▶ The closure risk factor (CRF) is simply a qualitative and quantitative measure that captures the various significant risk components of mine closure. The components are:
 - ▶ Environmental risks (RE): water, air, land, wastes etc.
 - ▶ Safety and health risk (RSH): openings, subsidence, infrastructures, security etc.
 - ▶ Community and social risk (RC): employees, management, unions, landowners, community impact etc.
 - ▶ Final land use risks (RLU): : high value , premium agriculture land
 - ▶ Legal and financial risks (RLF): Government, security bonds, taxes, salvage etc.
 - ▶ Technical risk(RT): plan exists and up to date, planning, resource/reserves, etc.

To calculate the CRF weightage is given to primary risks and the secondary risks and simply adding them to get CRF. It is varying from 2500 to more than 10,000 so higher the CRF higher the risk of closure.

Conclusions

- ▶ There are basically two main aspects of mine closure planning - firstly, environmental stabilization and secondly, the economic consequences. Among these, economic effect demands judicial planning with especial consideration of the sentiments of the local residents. It is evident that alternate resource generation in terms of cultivation is almost impossible. In fact, post mine land profile permits set up neither of industries nor of other long term resource generation activities. This leads to migration of the local residents in search of food. A common solution to this problem can not be established. However, a site-specific guideline should have been established for the mines in future.
- ▶ Mine closure planning has to be done at the starting point of the mining operations and needs periodic review and revision during its life cycle to cope with the geo- technical constraints, safety and economic risks, social & environmental challenges.

